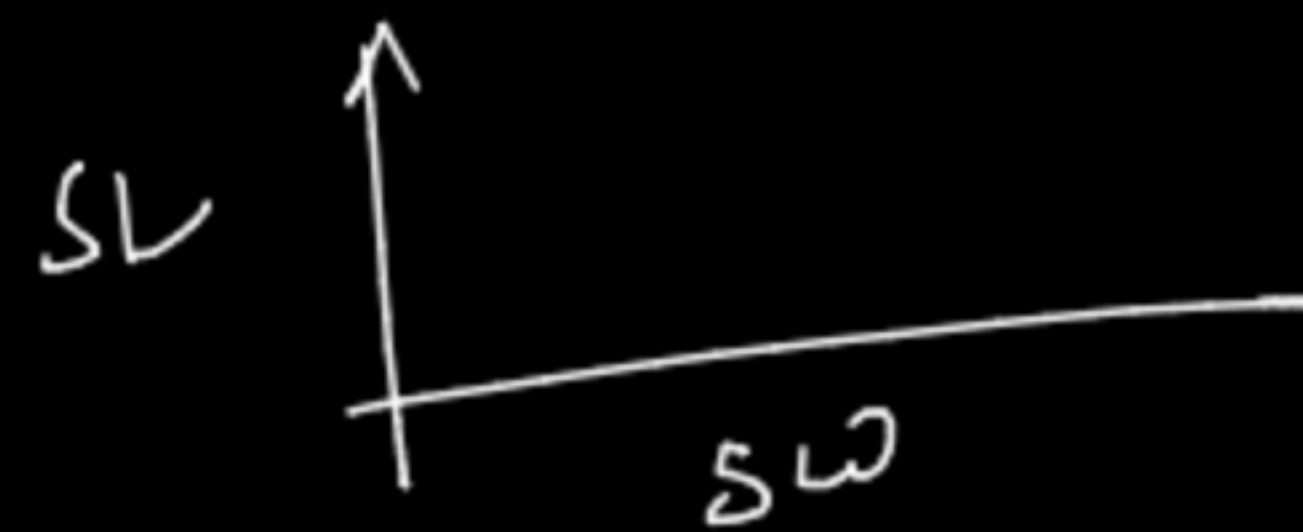
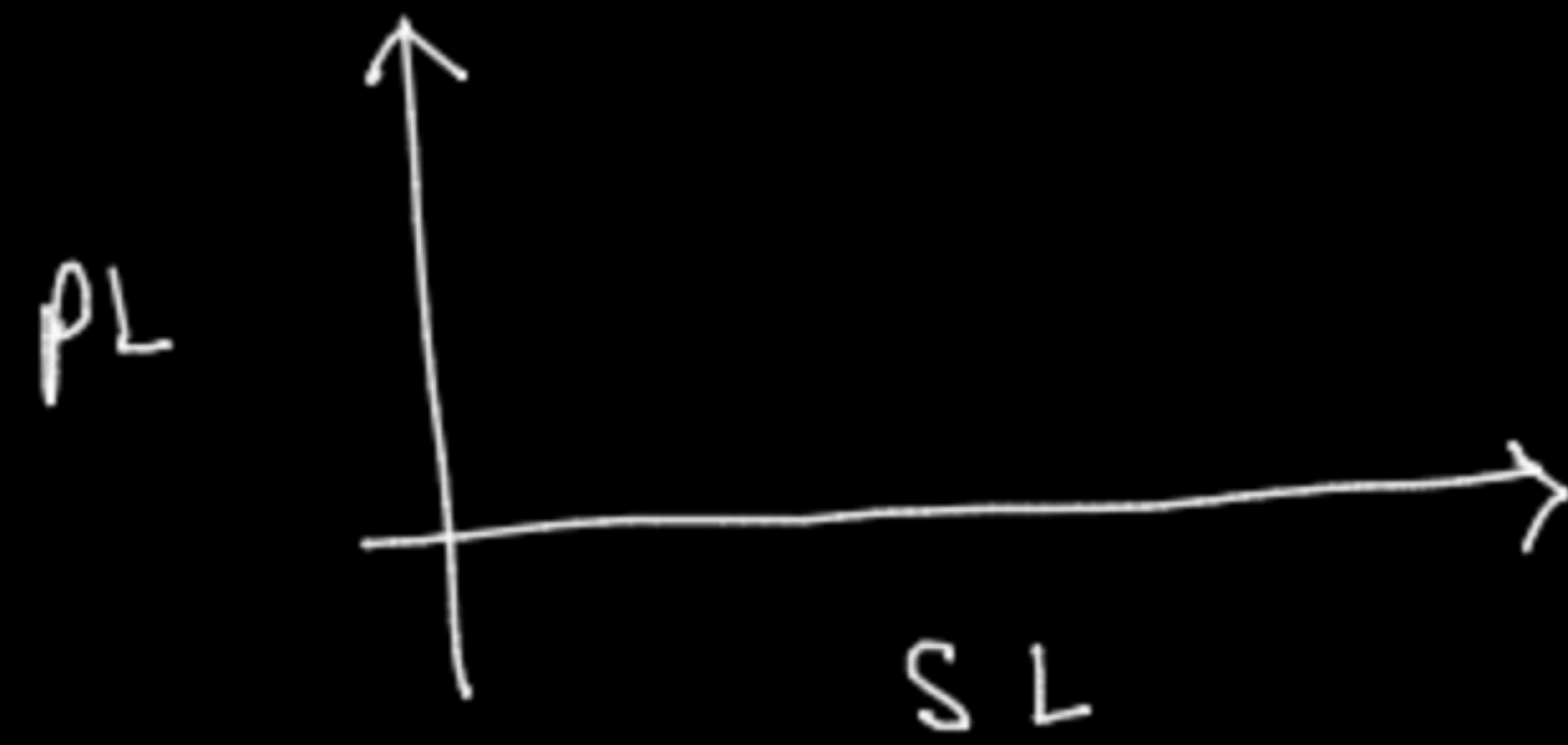


LA → ?
= =

SL SW PL PW species
- - - -

↓ ↓



path-plot → [2D]
↑

Rule engine

↑

↓

[if PL > and SL < else else else]

PL

SL

if

[3D]



if $SL > 2.5$ and $PL < 1.5$

if $pw < 0.2$ and $sw > 0.15$

return $\rightarrow$ setos4

if $(PL < 2.5$ and $SL < 0.5)$ and $pw < 0.7$
 $\rightarrow$ setos5

Rule1

inis $\rightarrow$ Linear Algebra

conditions

2D



3D



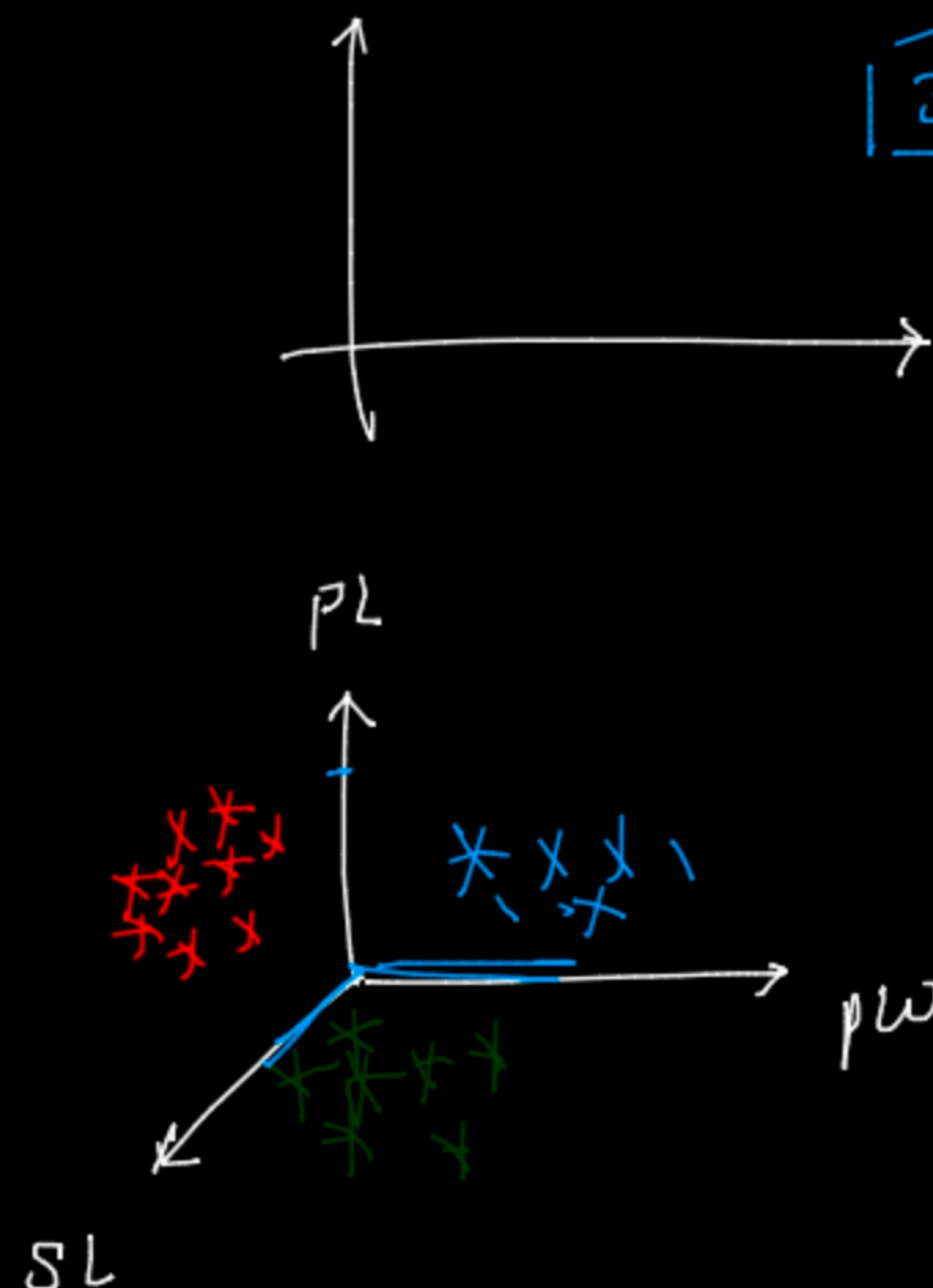
But

∂_x

4

5

n

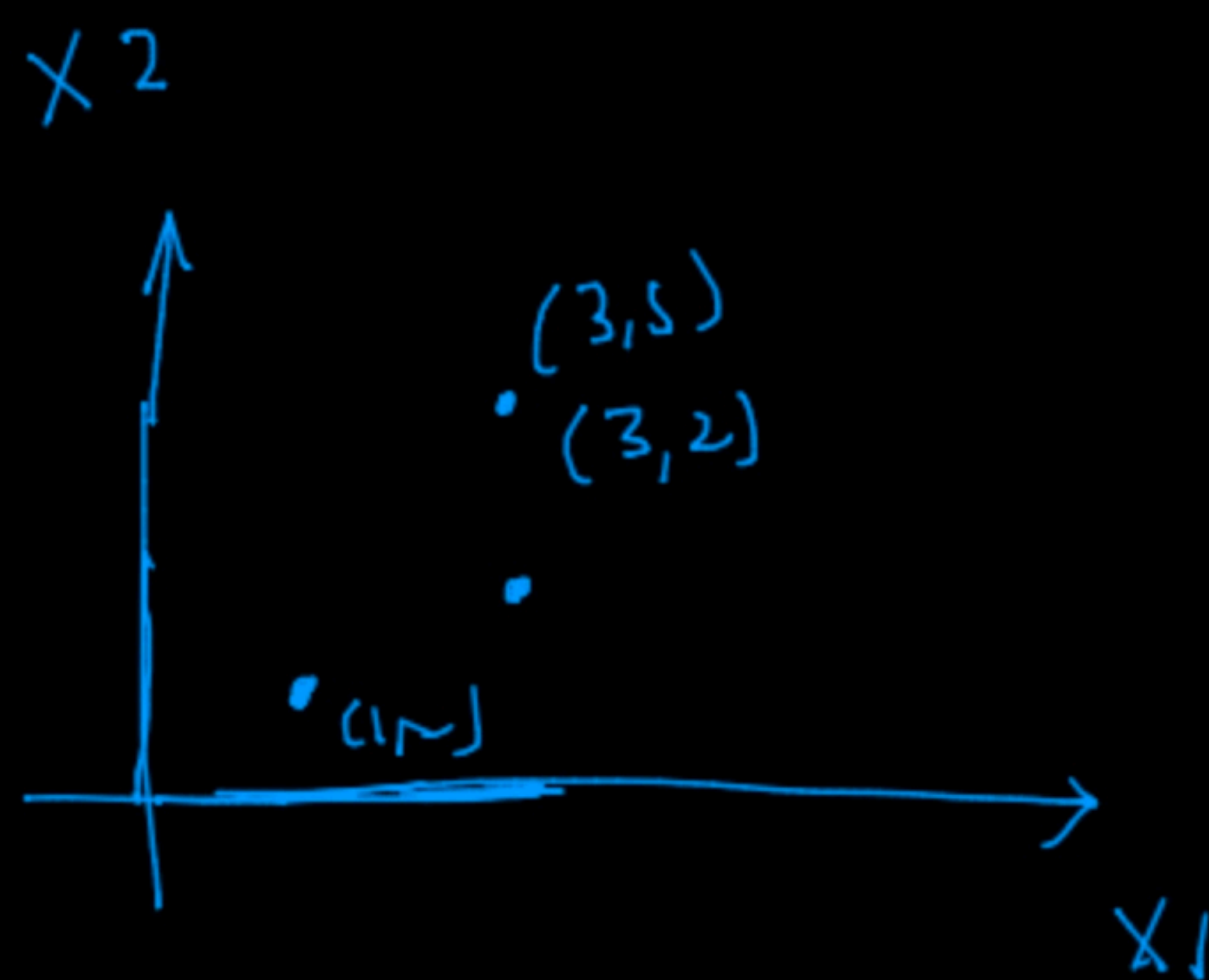


17 Points $\rightarrow$ vectors

(case i) $\uparrow$

	x_1	x_2
vector $\rightarrow$	(3)	(2)
	(1)	(2)
	(3)	(5)

$\rightarrow$



(case II)

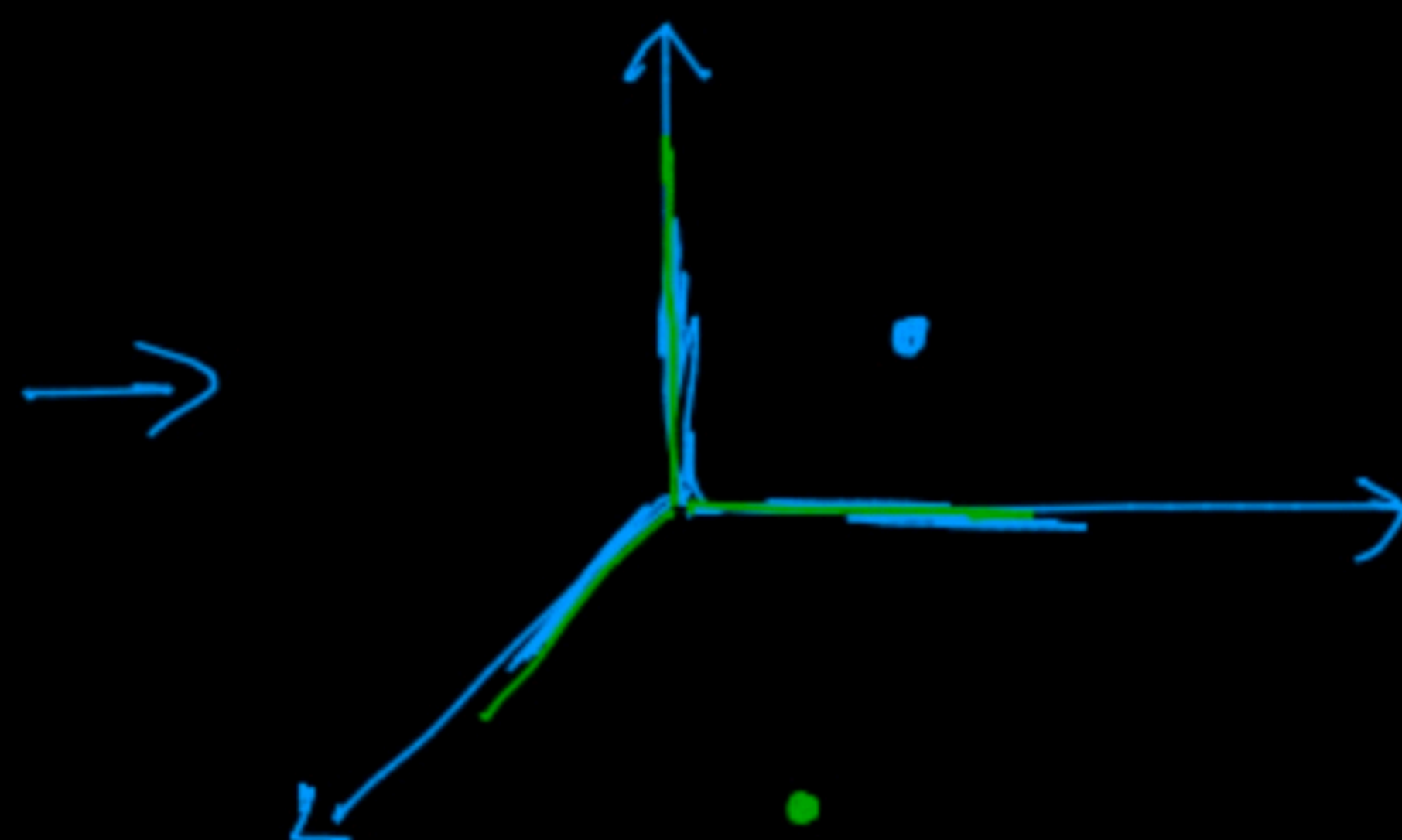


	x_1	x_2	x_3	x_4
$\rightarrow$	4	5	2	1
	3	2	1	5
	6	1	2	5

(case II)

	x_1	x_2	x_3
$\rightarrow$	3	2	4
	4	5	6
	9	8	5

$\rightarrow$



$\Downarrow$
X

can't plot

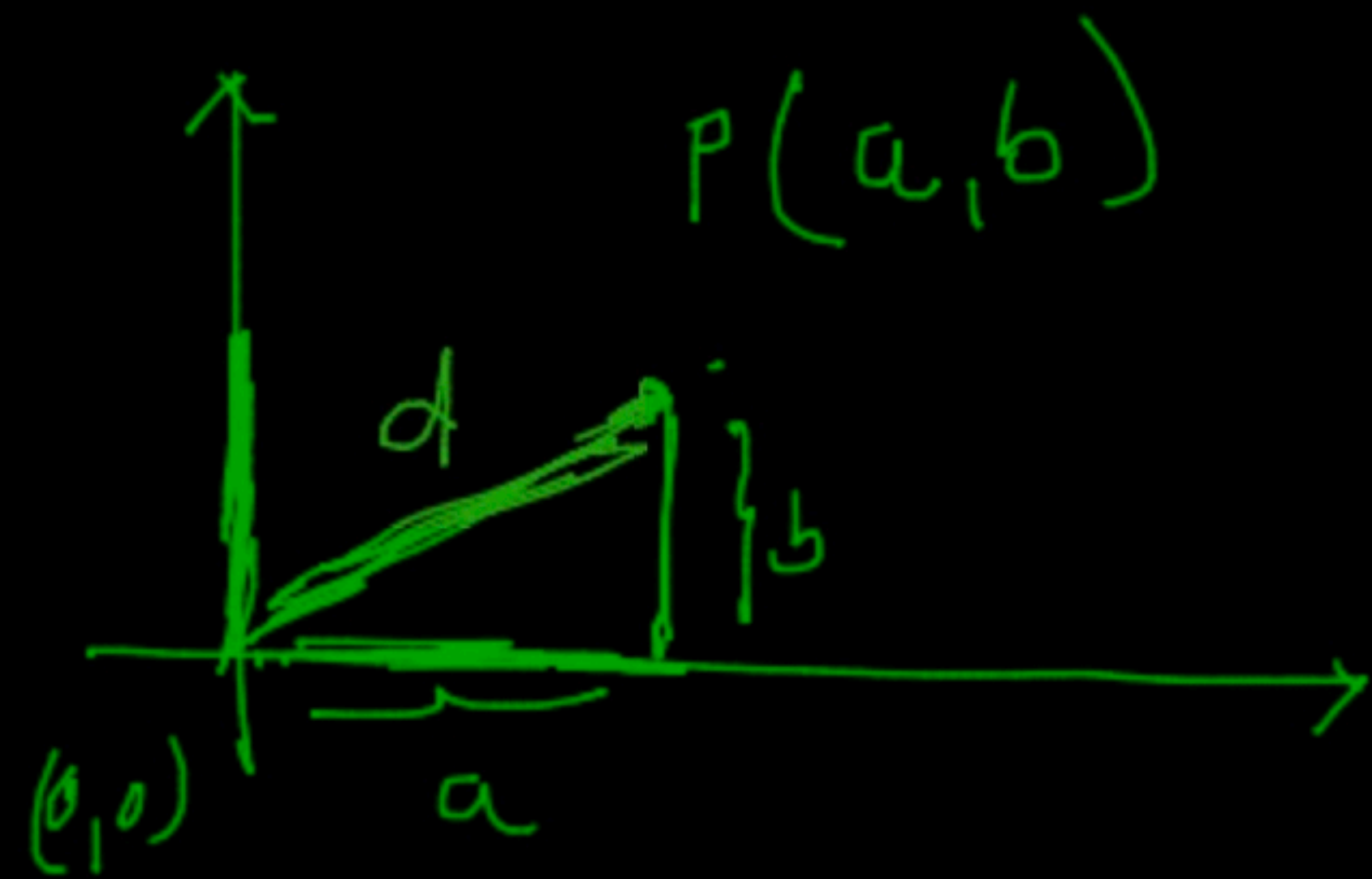
$\uparrow$
 u_1, u_2
 $\Rightarrow$

4d 5d --- no



(3) Distance of a points from origin

2D

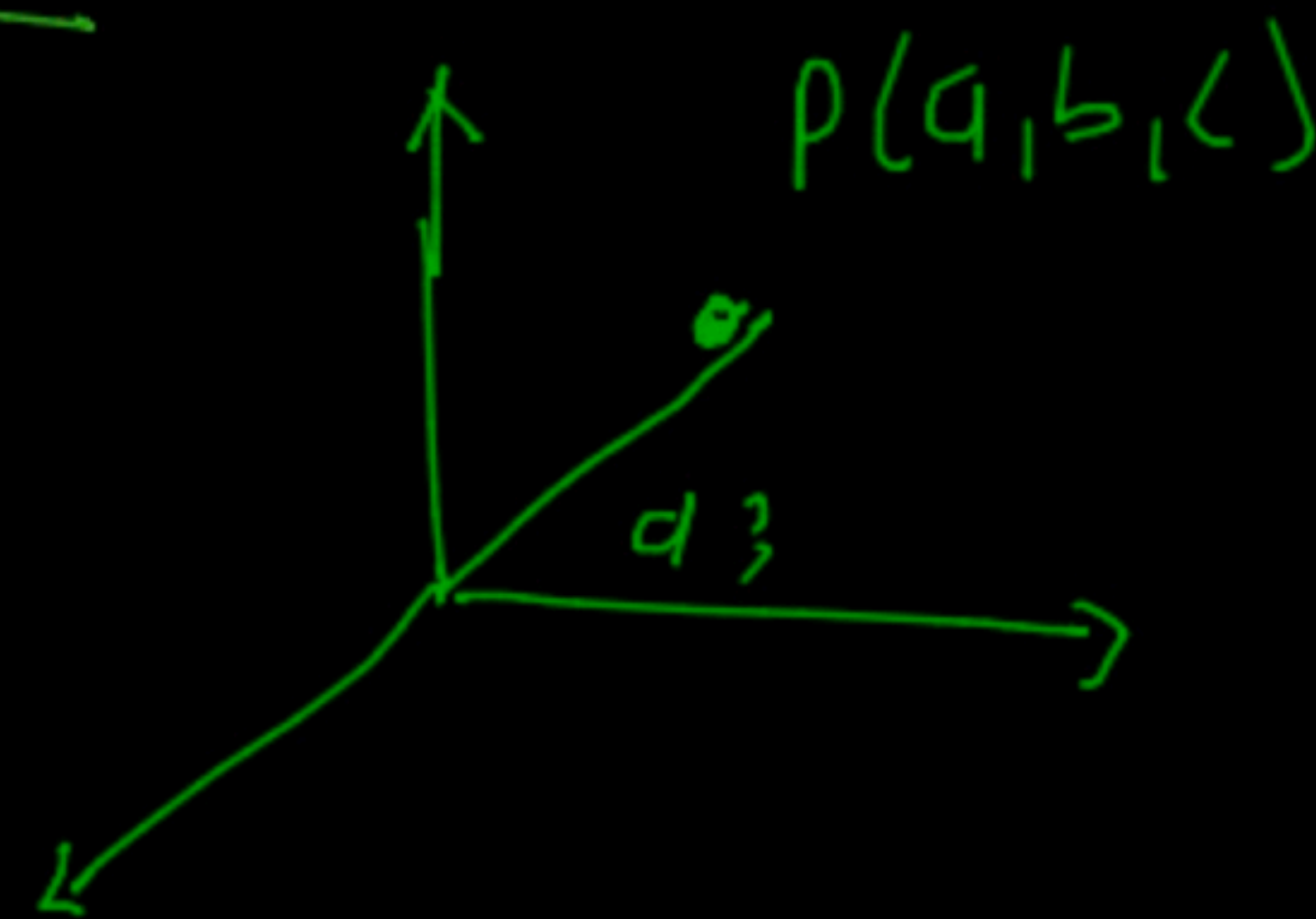


$$d^2 = a^2 + b^2$$

$$d = \sqrt{a^2 + b^2}$$

Imp?

3D



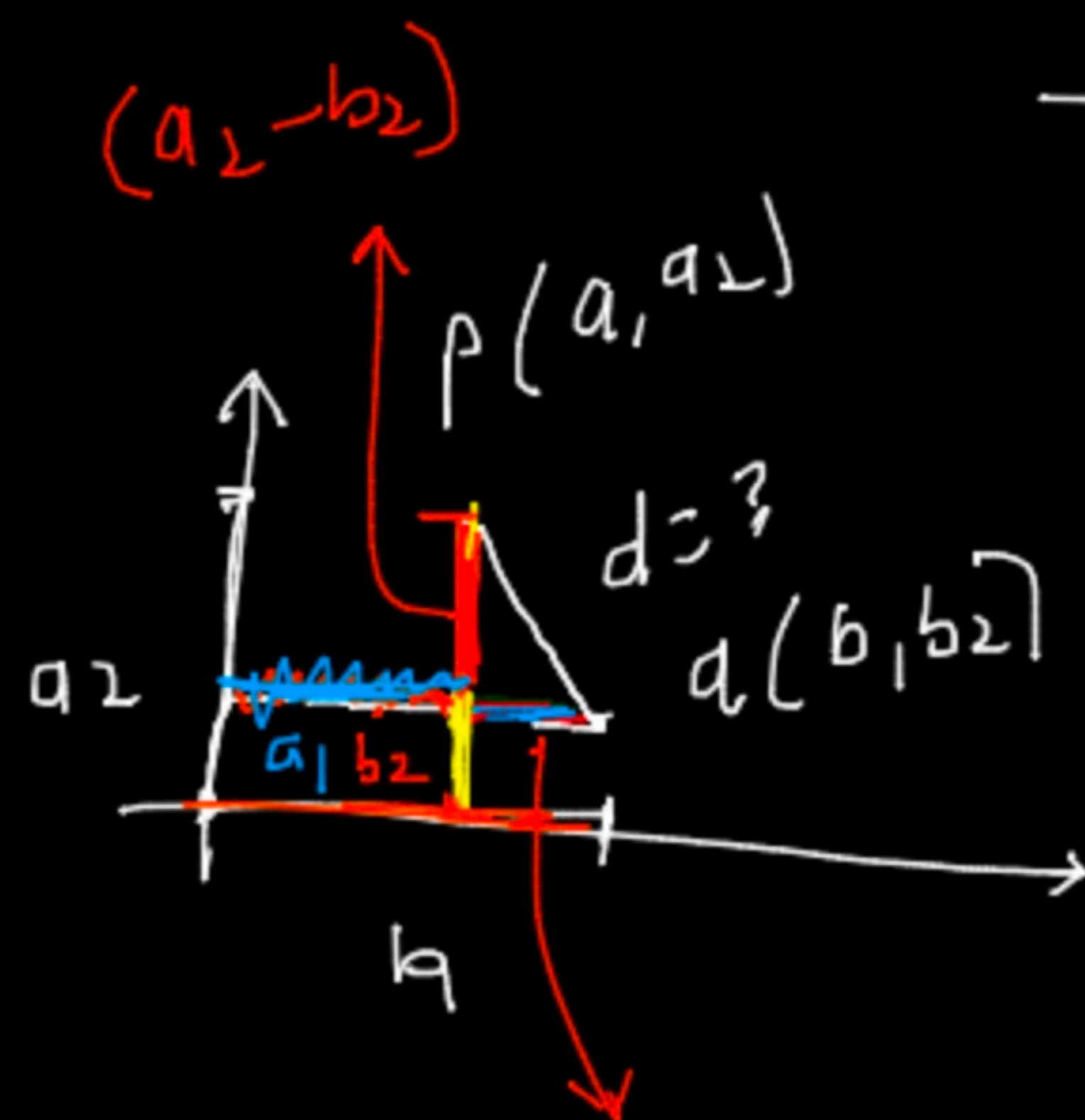
$$d = \sqrt{a^2 + b^2 + c^2}$$

'nD' $\rightarrow P(a_1, a_2, a_3, a_4, \dots, a_n)$

$$d = \sqrt{a_1^2 + a_2^2 + a_3^2 + a_4^2 + a_5^2 + \dots + a_n^2}$$

⇒ distance b/w 2 points

12D



→ 3D

$P(a_1, a_2, a_3)$ $Q(b_1, b_2, b_3)$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$

nD

$P(a_1, a_2, a_3, \dots, a_n)$ $Q(b_1, b_2, b_3, \dots, b_n)$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 + \dots + (a_n - b_n)^2}$$

↓
short-cut

$$d = \text{sqrt}\left(\sum_{i=1}^n (a_i - b_i)^2\right) \leftarrow \text{form}$$

↓
code

$P(a_1, a_2)$
 $Q(b_1, b_2)$

$$d = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

eg,

$$C_1 = \begin{pmatrix} x_1 \\ 17, 750 \end{pmatrix}$$

$$C_2 = \begin{pmatrix} x_2 \\ 16, 750 \end{pmatrix}$$

$$\Rightarrow \text{distance} = \sqrt{(1)^2 + (0)^2}$$

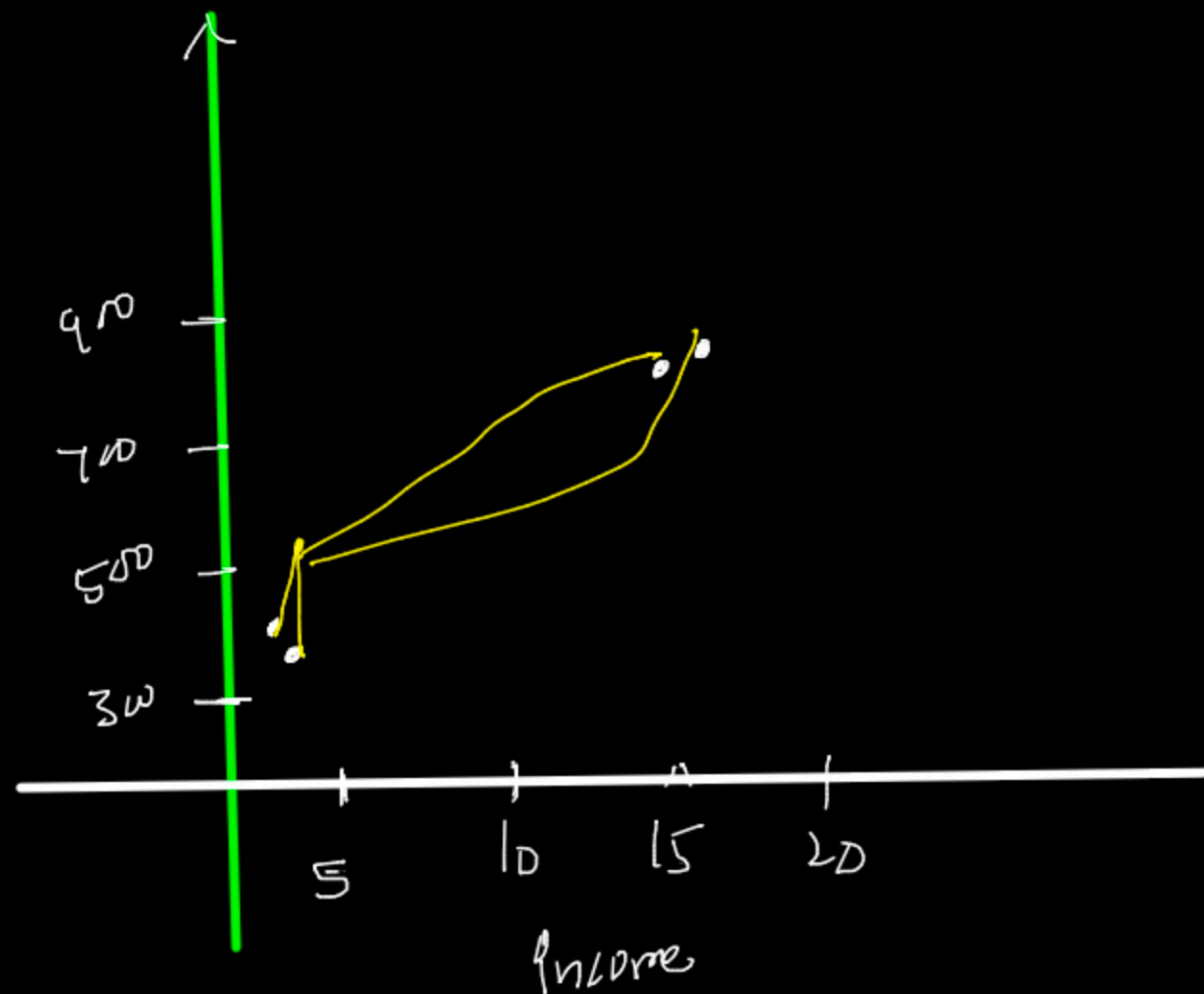
1

eg.

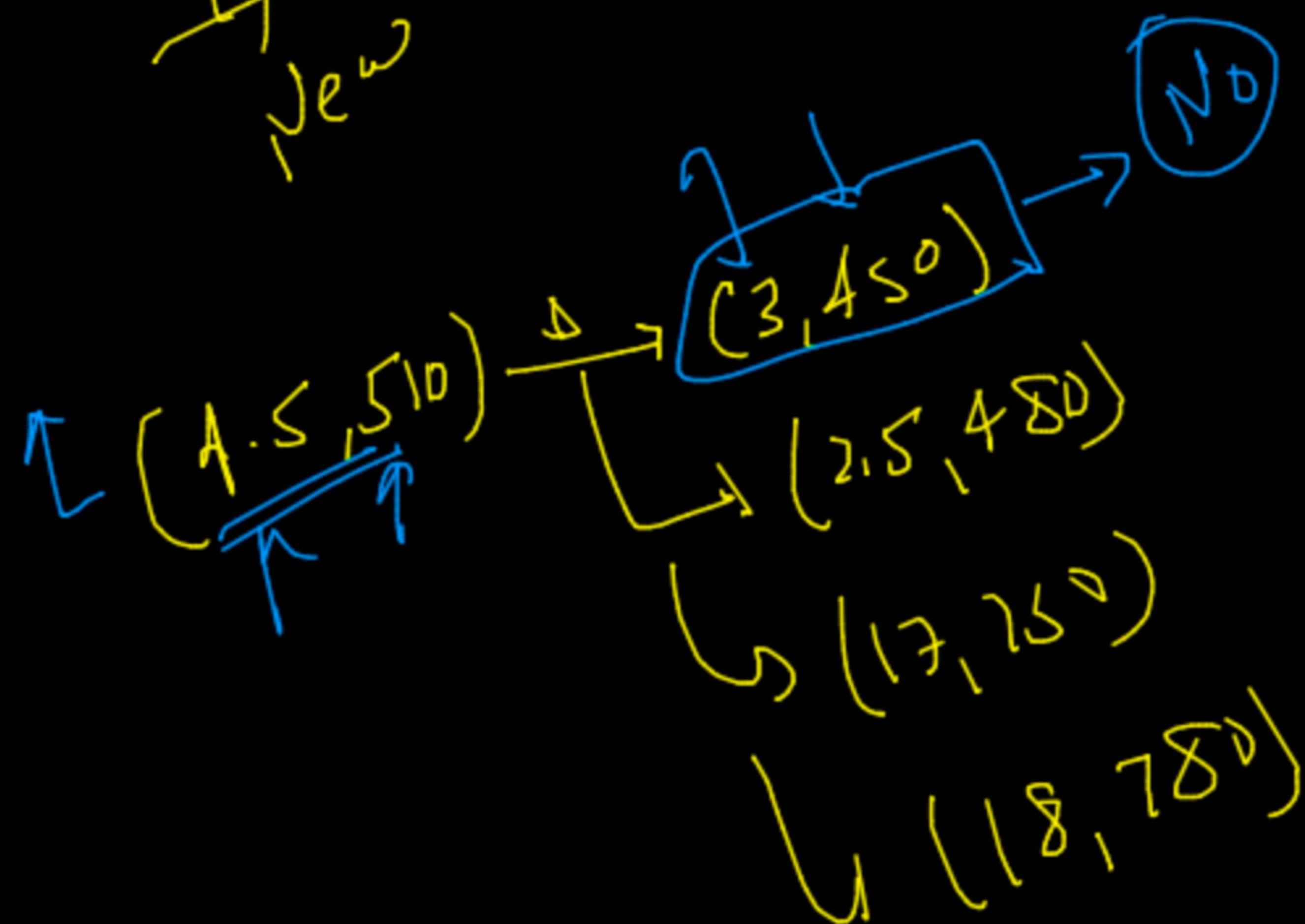
	Income	CS	Status
	(3	450)	No
	2.5	480	No
	17	750	Yes
	18	780	No
	(A.5	510)	→ Yes/No

Plot

Income
CS



(A.5)



Algorithm → KNN

Row vector $\rightarrow A = [a_1 a_2 a_3 \dots a_n]$

Col vector

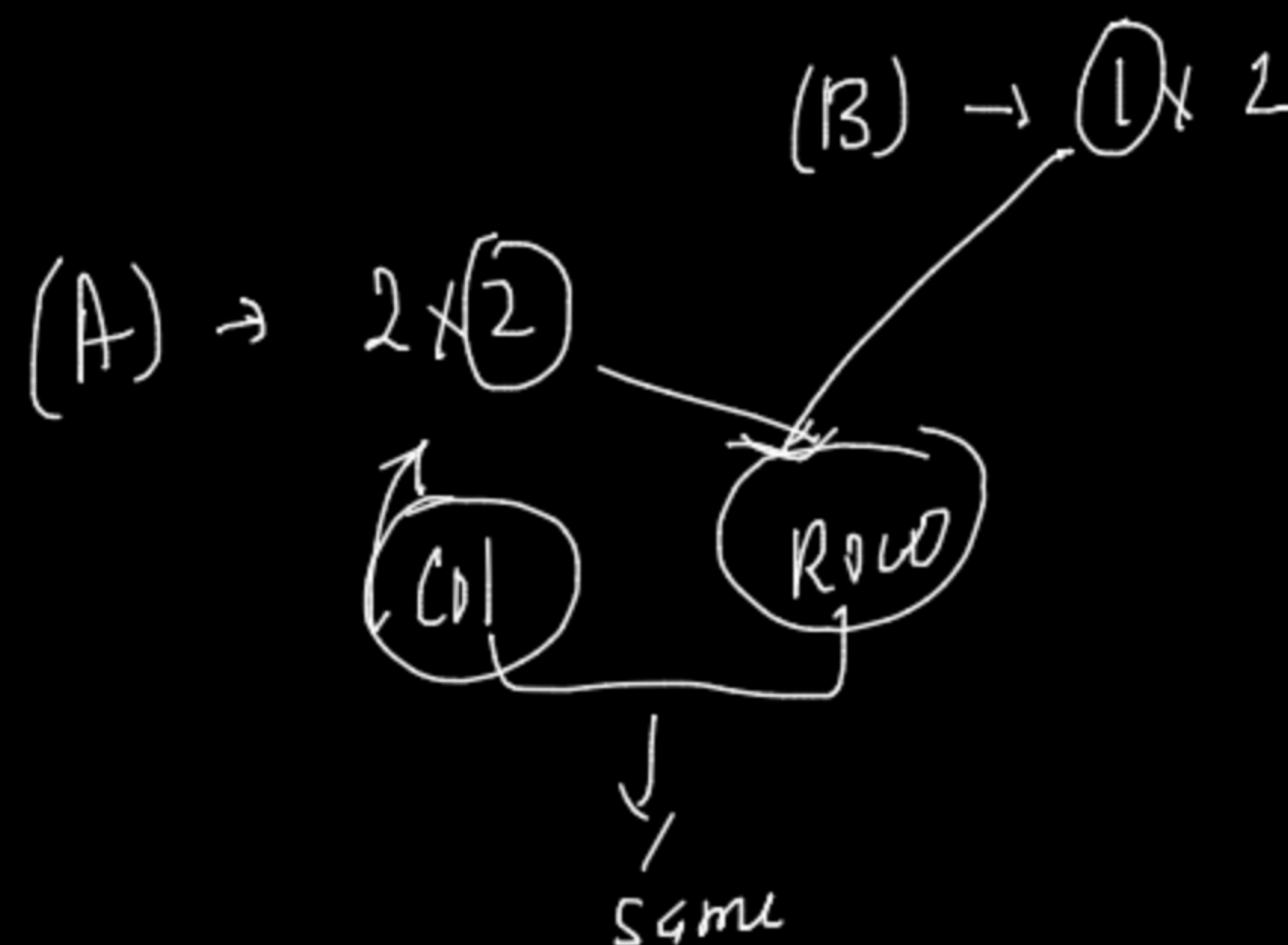
$$A^T = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{bmatrix}$$

eg- $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 1 & 2 \end{bmatrix}$

$$B^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

(2×2) $\swarrow$ same A (2×1)



$$a = [a_1 \ a_2 \ a_3 \ \dots \ a_n]$$

$$b = [b_1 \ b_2 \ b_3 \ \dots \ b_n]$$

eg. $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

1) $(a+b) \rightarrow$ component wise addition

2) $\underline{a \times b} \rightarrow$ dot product $\square$
 $\hookrightarrow$ cross product $\times$ (Not in ML)

$$a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

$$[a_1 \ a_2 \ a_3 \ \dots \ a_n]$$

$$\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (n \times 1)$$

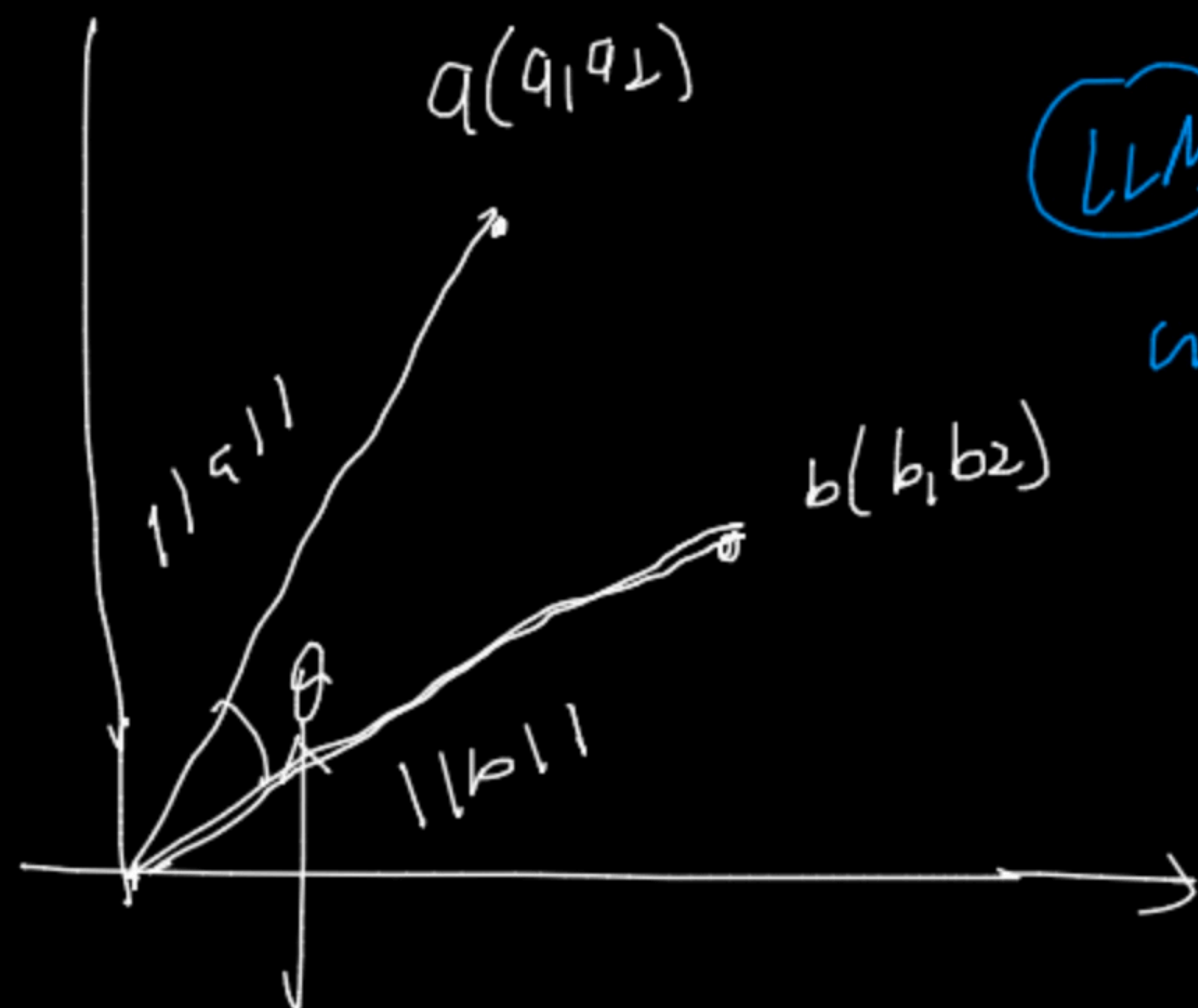
$$\rightarrow [\]_{1 \times 1}$$

$$1 \times n$$

$$\sum_{i=1}^n a_i \cdot b_i$$

Geometrically mean

[2D]

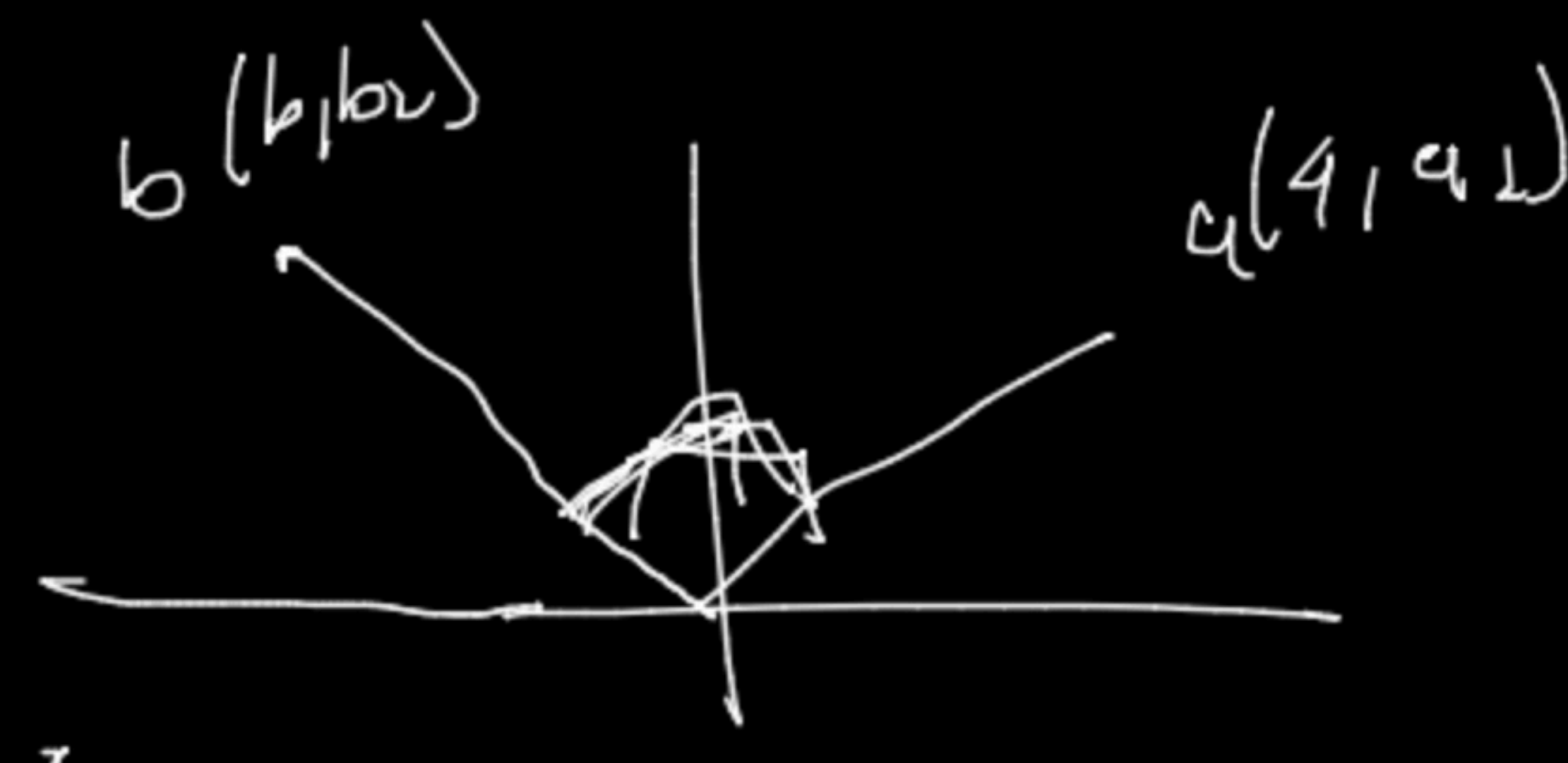


cosine-similarity

LLM $\rightarrow$ 'RAG'

$\theta \downarrow \uparrow$

eq $\theta = \angle$



$$a \cdot b = \|a\| \|b\| \cos \theta$$

$$a \cdot b \geq 0$$

$$a \cdot b = \underbrace{\|a\| \|b\|}_{\text{length of } a} \cdot \cos \theta \rightarrow$$

length of a

$$a_1 b_1 + a_2 b_2 = \|a\| \|b\| \cdot \cos \theta$$

$$\theta = \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|} \right)$$

C1

C1

C6



C1

C3

C4

